To What Extent Does Secondary School Enrollment Predict Youth Unemployment?

Decision analysis | World Bank Data | 9 countries | 2000–2023

PYTHON → SQL → POWER BI

SUB-SAHARAN AFRICA & MAGHREB

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Project Overview

Business context

Youth unemployment is a major socioeconomic challenge in Sub-Saharan Africa and the Maghreb. Governments are investing heavily in secondary education with the assumption that higher enrollment reduces unemployment. This hypothesis deserves to be tested on real-world data.

Analytical objective

Measure the relationship between secondary enrollment rates, youth unemployment, and education spending (2000–2023) to inform public policy decisions.

Manually built dataset

Since the World Bank site does not provide a single consolidated file, each indicator was downloaded separately and then merged in Python.

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Dataset & Analytical Pipeline

Dataset dimensions

18 448

Raw rows

5 raw columns → 11 columns after enrichment

9

Countries covered

Sub-Saharan Africa + the Maghreb

23

Analysis years

Period 2000–2023

Key variables

Enrollment

taux_scolarisation_secondaire (%)

Unemployment

taux_chomage_jeunes (%)

Spending

depenses_education_pib (% of GDP)

Analytical pipeline

Python (pandas)

Cleaning, feature engineering, CSV export

SQL MySQL Workbench

10 decision-making analytical queries

Power BI

3-page decision-support dashboard

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Quality Audit & Feature Engineering

Applied treatments

→ **Missing values**

Country median + global median

→ **Outlier values**

IQR capping — robust to socioeconomic asymmetries

→ **Unusable rows**

Removal with dropna how="all"

5 business variables created

Variable	Rule	Use
ratio_efficacite	unemployment / GDP expenses	Investment efficiency KPI
segment_scolarisation	Low / Medium / High	Power BI slicer
segment_chomage	Low / Moderate / Critical	Decision alert
segment_effort_education	Low / Medium / High	Budget comparison
variation_chomage	Year N vs N-1 difference	Country trend

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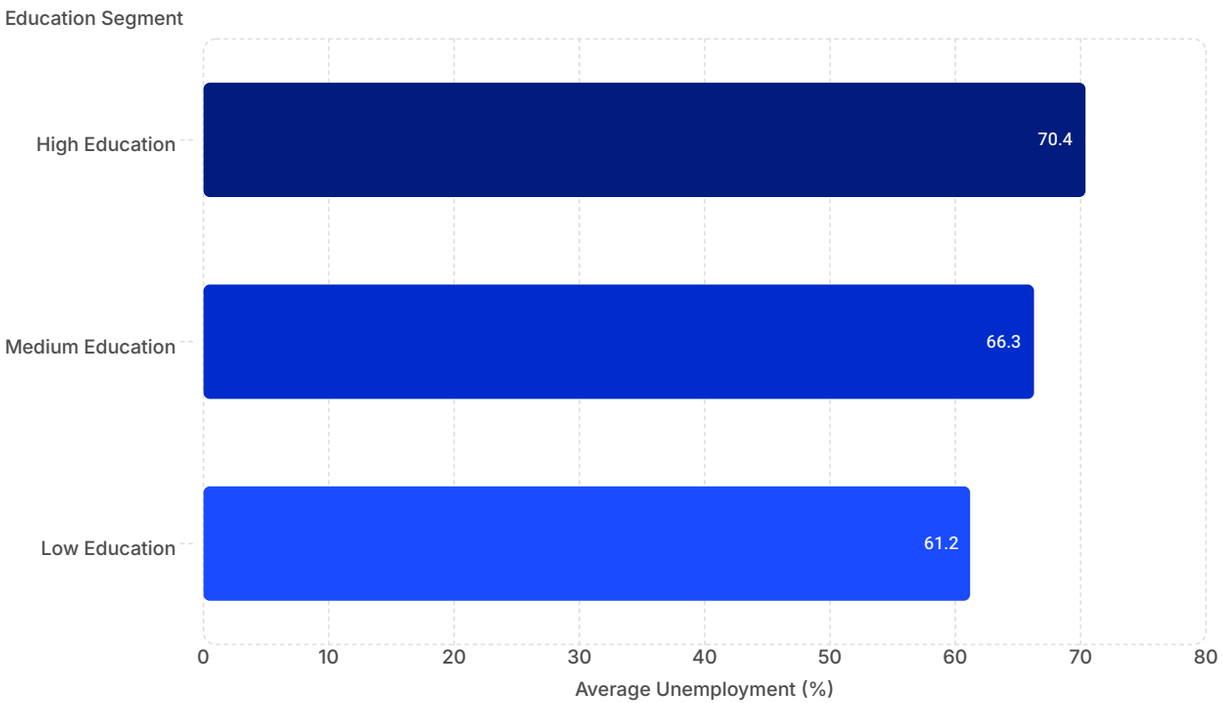
Key Result #1: Education does not mechanically reduce unemployment



Decision insight: Counterintuitively, countries with high education levels show the highest youth unemployment. Education alone is not enough — alignment between training and employment is essential.

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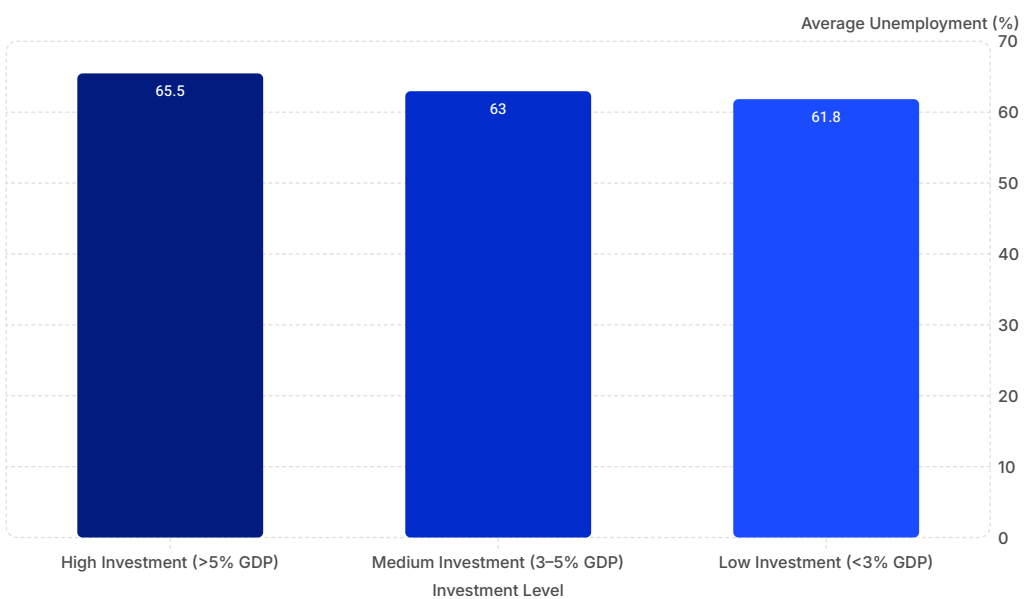
Key Result #2: Educational investment is not a direct lever



Decision insight

The countries that spend the most on education do not achieve the best employment outcomes.

Spending efficiency matters more than spending volume.

This result calls into question education funding policies based solely on budget size, without measuring their impact on job placement.

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Key Result #3: Countries with a Deteriorated Efficiency Ratio

High unemployment despite significant investment — the priority countries for intervention.

Country	Efficiency Ratio	Average Unemployment	Spending as % of GDP	Priority
Nigeria	**32.74**	62.85%	1.92%	🔴 Critical
Tanzania	16.38	62.48%	3.88%	🟠 High
Egypt	14.31	66.04%	4.62%	🟠 High
Senegal	13.87	56.25%	4.53%	🟡 Moderate
Morocco	13.45	69.63%	5.19%	🟡 Moderate

⚠️ **Nigeria Alert:** Nigeria generates **32.74 points of unemployment per point of GDP invested** — absolute priority for intervention. A full audit of education spending is urgently required.

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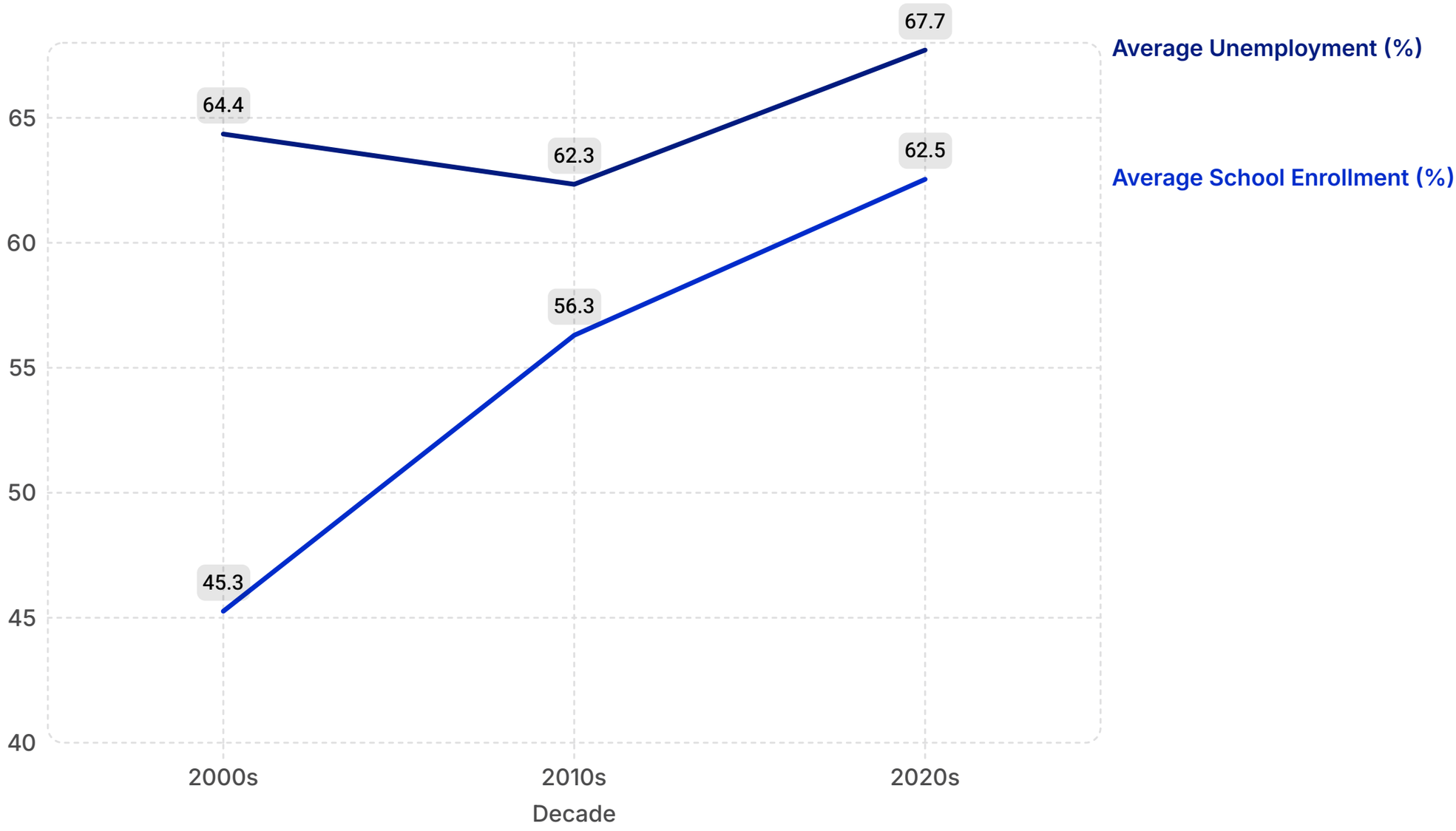
Key Result #4: Time Trend 2000–2023



Decision insight: Despite steady progress in school enrollment (+17 points over 20 years), youth unemployment is rising again in the 2020s. The pandemic and economic shocks erased the structural gains.

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Key Result #5: 4 Benchmark Countries Identified

These countries combine high schooling rates with contained unemployment — models to study and transfer.

🇪🇬 Egypt

Schooling: **83.23%**

Unemployment: **5.48%**

Efficiency ratio: 1.17

🇰🇪 Kenya

Schooling: **82.73%**

Unemployment: **5.59%**

Efficiency ratio: 1.26

🇹🇳 Tunisia

Schooling: **87.16%**

Unemployment: **6.04%**

Efficiency ratio: 0.94

🇲🇦 Morocco

Schooling: **83.04%**

Unemployment: **6.11%**

Efficiency ratio: 1.03

These 4 countries show that the combination of **high schooling + active employment policies** produces results. Their practices deserve to be documented and transferred to countries facing difficulties.

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Business Recommendations & Conclusion

5 Decision-Making Recommendations

1	**Reorient education policy** Align secondary curricula with real labor market needs
2	**Audit budget efficiency** Nigeria, Tanzania, Egypt → Targeted spending reviews focused on employment outcomes
3	**Transfer best practices** Cross-country exchange missions: Morocco, Tunisia, Kenya, Egypt
4	**Complement educational investment** Active labor market policies → apprenticeships, job placement, youth hiring subsidies
5	**Condition education budgets** Placement rate as a funding KPI — link budgets to employment outcomes

What the analysis demonstrates

Secondary school enrollment is not a direct predictor of youth unemployment. The relationship is complex, non-linear, and strongly depends on how well education aligns with labor market needs.

Explicit limitations

- Uneven World Bank country coverage over the long term
- Missing values filled with the median → slight homogenization
- Enrollment ≠ completed diploma (quality not measured)
- Correlational, not causal — no control variables

Complete pipeline

Python (EDA + cleaning) → SQL MySQL Workbench (10 queries) → Power BI (3 pages) → Report + Presentation

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